

Kenneth Zhang

UNDERGRADUATE RESEARCHER | LEARNING-AUGMENTED OPTIMIZATION

kennethzhang29@outlook.com

github.com/KennethZhang29

Guangzhou | Macau | Zhuhai

PROFILE

Undergraduate researcher at the Faculty of Data Science, City University of Macau. I study how learned structural priors and group-aware policies can guide classical optimization solvers for constrained routing and intelligent transportation.

12 / 12

portfolio-best matches

3.10x

lower selection cost

2026

iLSCA Lab research assistant

RESEARCH EXPERIENCE

Research Assistant, iLSCA Lab

JUN-AUG 2026

The Hong Kong University of Science and Technology (Guangzhou)

- Developed and evaluated learning-guided methods for constrained routing, connecting neural structural signals with LKH-style heuristic search.
- Built reproducible benchmark, ablation, diagnostic, and scientific-visualization workflows.

SELECTED RESEARCH

Learned Edge Priors for Population-LKH

FIRST-AUTHOR PROJECT

- Designed a two-stage framework with cluster-aware candidate construction, Transformer edge priors, adaptive sparsification, and guarded solver selection.
- On 12 finite-L BWTSP instances, matched the portfolio-best choice on 12/12 cases while reducing selection cost by about 3.10x.

GC-POMO for Group-Constrained TSPs

CO-AUTHOR

- Contributed software and research visualization for a group-aware attention encoder and interactive dual-decoder spanning BWTSP and CTSP-d.
- The 3,000-rollout model improved over POMO on BWTSP50/100 and CTSP-d20/50/100, with stronger cross-size generalization.

Energy-Constrained Delivery

ONGOING

- Exploring payload-dependent routing, optional charging stations, battery limits, and feasibility-aware guidance for drone and electric-vehicle delivery.

RESEARCH WORKFLOW

- Formulate operational constraints and construct reproducible benchmark instances.
- Learn edge, group, move, or route-state signals without replacing reliable solver mechanics.
- Integrate guidance through native solver interfaces and feasibility-aware action masks.
- Audit claims with strong baselines, ablations, multi-seed tests, and route-level diagnostics.

EDUCATION

City University of Macau

Faculty of Data Science

B.Eng. in Computer Science and Technology

2023-2027 (expected)

TECHNICAL SKILLS

OPTIMIZATION

LKH3, Gurobi, ALNS, local search

LEARNING

PyTorch, Transformer, attention, GNN, policy gradient, POMO

ENGINEERING

Python, C++, Linux, Git, CUDA

RESEARCH

Baseline reproduction, ablation, benchmarking, scientific visualization

RESEARCH INTERESTS

Learning-Augmented Optimization
Combinatorial Optimization
Evolutionary Computation
Reinforcement Learning
Vehicle Routing and Energy Delivery

LANGUAGES

Mandarin and Cantonese: Native
English: Professional working proficiency

BEYOND RESEARCH

Football, tennis, film, and CS2. Chelsea supporter.

SELECTED COURSEWORK

Data Structures and Algorithms
Optimization Theory
Probability and Statistics
Reinforcement Learning
Deep Learning